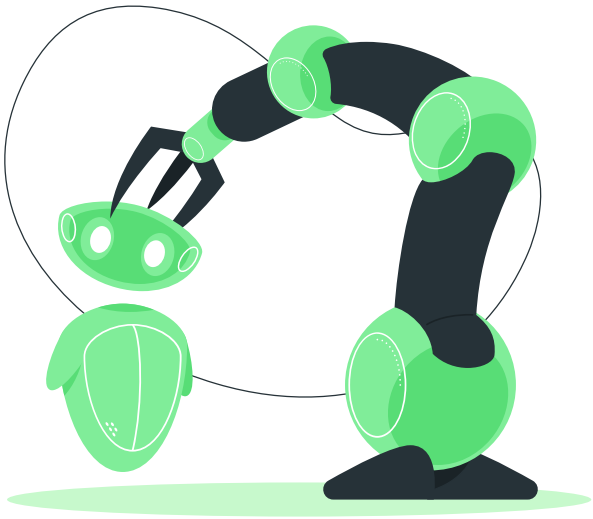
Robotics
CS 3015

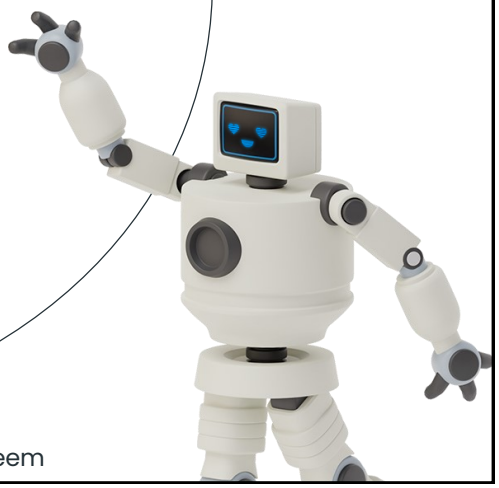
Umarah Qaseem



1

4

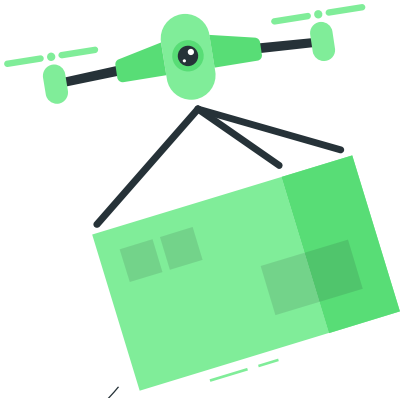
- Introduction to Robotic Arm
- Types of joints
- Types of robots based on joints
- 1 link serial manipulator
- 2 and 3 link Planar serial manipulator
- 3D Robot Arm
- Spaces



2

What did we learn?

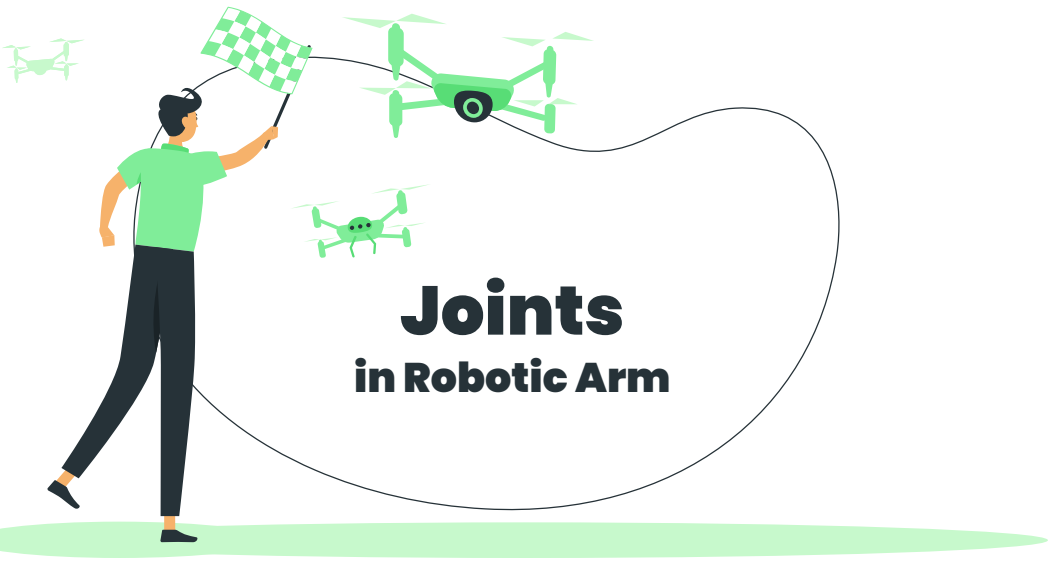
Discussion



Revision

Robotics – Umarah Qaseem

3



Joints
in Robotic Arm

Robotics – Umarah Qaseem

4

Types of Joints

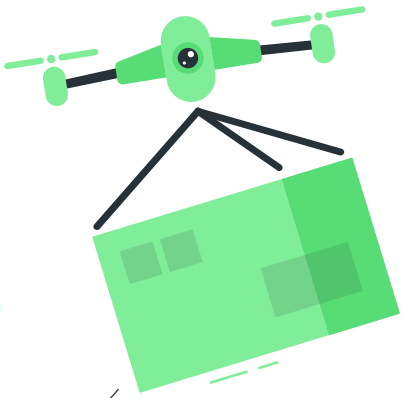
- Revolute
- Prismatic

Revolute Joint: Rotational movement around a fixed axis, 1 degree of freedom (rotation angle,
).
Prismatic Joint: Linear movement along a fixed axis, 1 degree of freedom (linear displacement,
d).

5

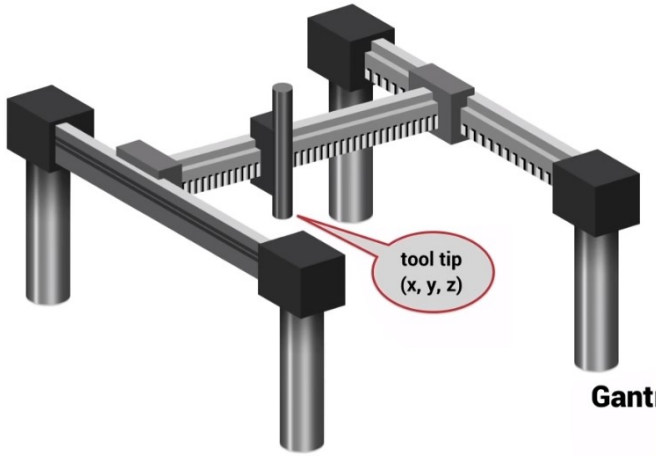
Types of Robots

With the perspective of joints



6

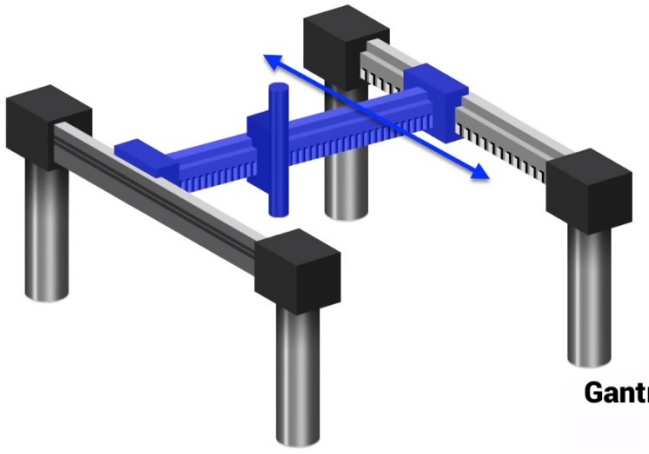
Gantry Robot



Gantry robot

7

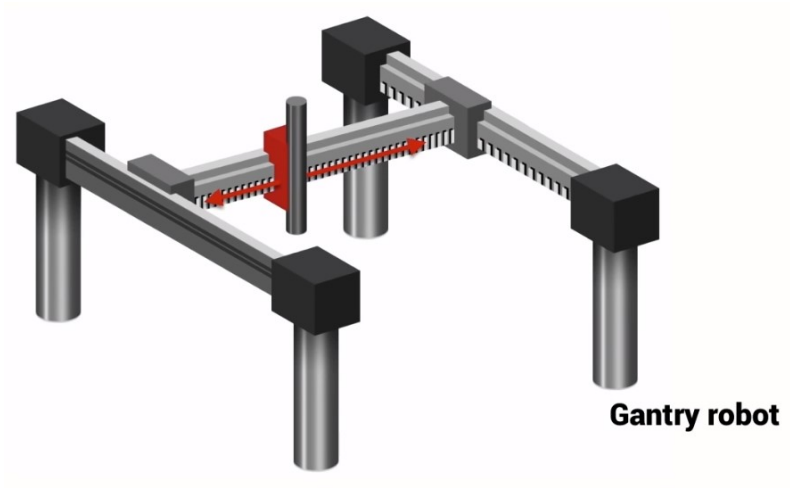
Gantry Robot



Gantry robot

8

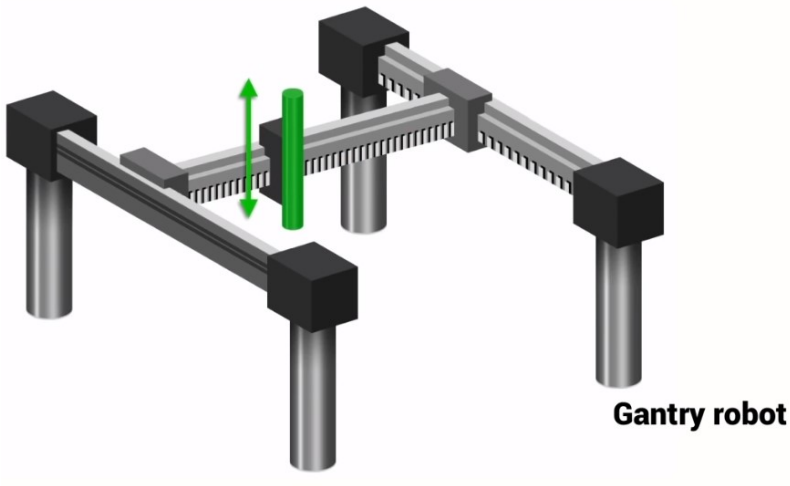
Gantry Robot



Gantry robot

9

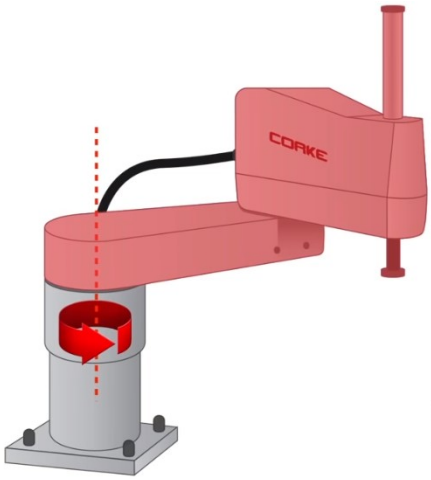
Gantry Robot



Gantry robot

10

Scara Robot

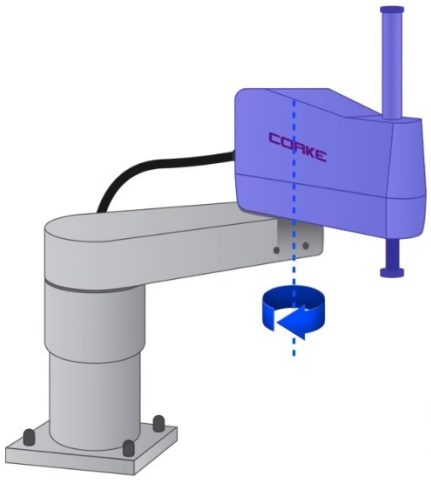


SCARA robot
1981

Robotics – Umarah Qaseem

11

Scara Robot

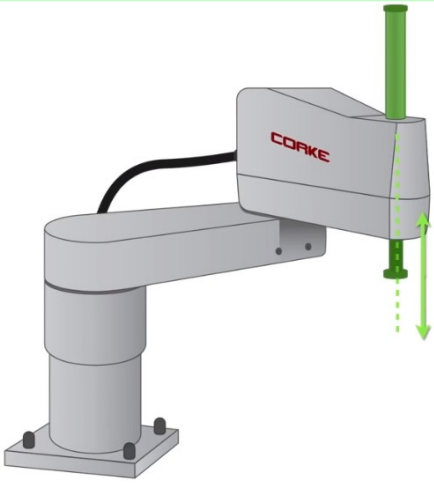


SCARA robot
1981

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Scara Robot



13

Scara Robot

Selective Compliance Articulating Robot
1981

Robot can move in horizontal plane but it's stiff in vertical plane

14

Parallel link robot



Robotics – Umarah Qaseem

15

Educational Robots



Robotics – Umarah Qaseem

16

Educational Robots



Robotics – Umarah Qaseem

17

Educational Robots – NAO

- 4 serial link manipulators – two arms and two legs
- Two at shoulder, one at elbow, one at wrist
- Wrist joint holds position and orientation of wrist and number of fingers
- Fingers themselves have further joints



Robotics – Umarah Qaseem

18

6 serial-link robot



19

6 serial-link robot

- Used to load and unload satellites and other cargo from the shuttle's payload bay.



Image: NASA

20

Stanford robotic arm

- 1969 in Stanford
- The year human walked on moon
- The third joint is sliding
- It changes distance with tool tip



Stanford robotic arm
Oussama Khatib | Used with permission

Unimate Robot

- First company to produce industrial robot
- 1978


Image: NASA Dominic Hart

A Famous Robot in Robotics History

- 1983
- Many many of them in manufacturing industry
- A large number of them in university research labs
- Many early robotics research papers are based on these

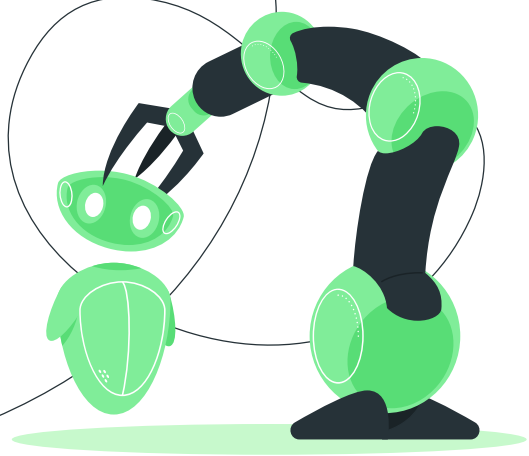
- **PUMA 560**



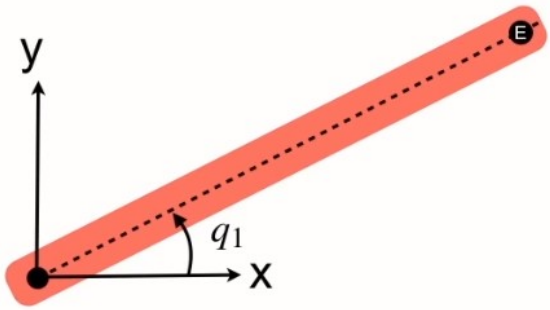
Unimate Puma 500
Oussama Khatib | Used with permission

1 joint Robot Arm

Analysis, Mathematics and derivation

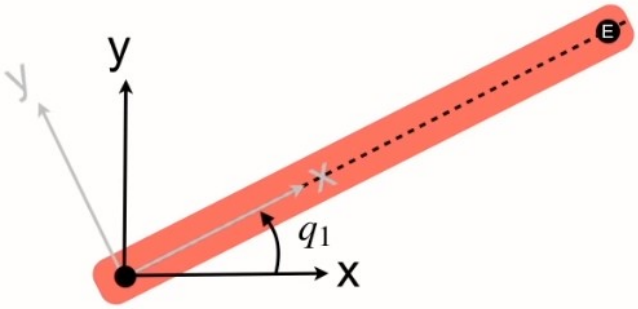


Tool tip pose



25

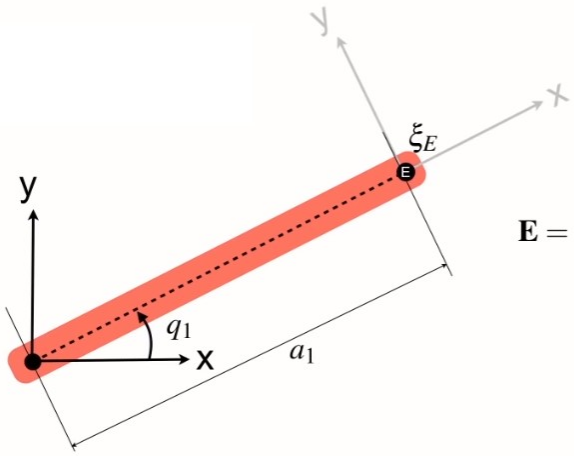
Tool tip pose



$$\mathbf{R}(q_1)$$

26

Tool tip pose



$$\mathbf{E} = \mathbf{R}(q_1)\,\mathbf{T}_x(a_1)$$

27

Tool tip pose

$$\mathbf{E} = \mathbf{R}(q_1)\,\mathbf{T}_x(a_1)$$

$$\mathbf{E} = \begin{pmatrix} \cos q_1 & -\sin q_1 & 0 \\ \sin q_1 & \cos q_1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & a_1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\mathbf{E} = \begin{pmatrix} \cos q_1 & -\sin q_1 & a_1 \cos q_1 \\ \sin q_1 & \cos q_1 & a_1 \sin q_1 \\ 0 & 0 & 1 \end{pmatrix}$$

28

Tool tip pose

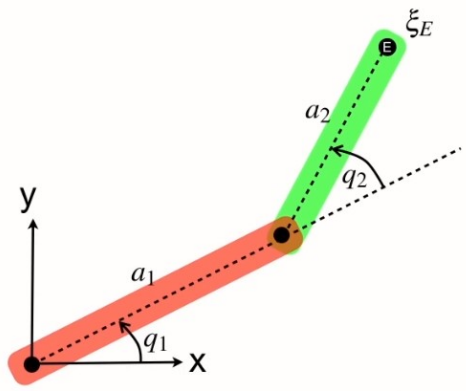
$$(x, y) = (a_1 \cos q_1, a_1 \sin q_1)$$

$$\mathbf{E} = \mathbf{R}(q_1)\ \mathbf{T}_x(a_1)$$

$$\mathbf{E} = \begin{pmatrix} \cos q_1 & -\sin q_1 & 0 \\ \sin q_1 & \cos q_1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & a_1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

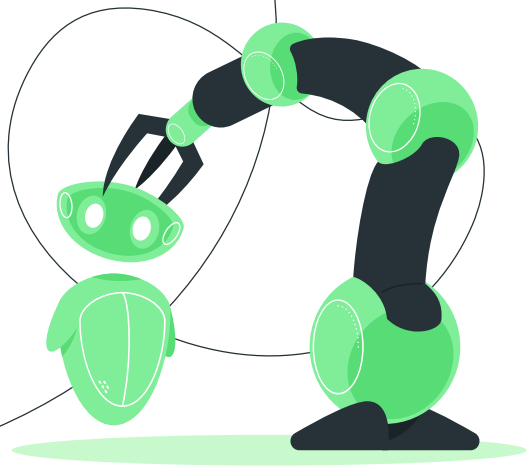
$$\mathbf{E} = \begin{pmatrix} \cos q_1 & -\sin q_1 & a_1 \cos q_1 \\ \sin q_1 & \cos q_1 & a_1 \sin q_1 \\ 0 & 0 & 1 \end{pmatrix}$$

2 joint Robot Arm



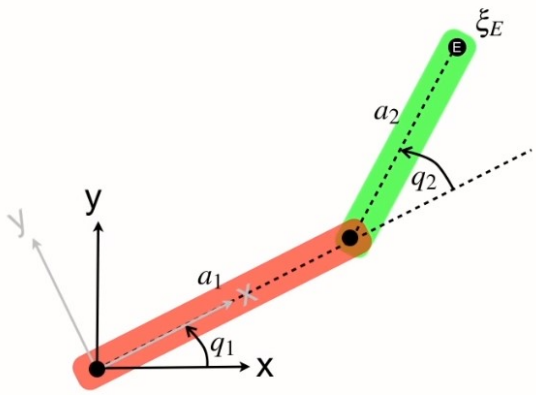
2 joint Robot Arm

Analysis, Mathematics and derivation



31

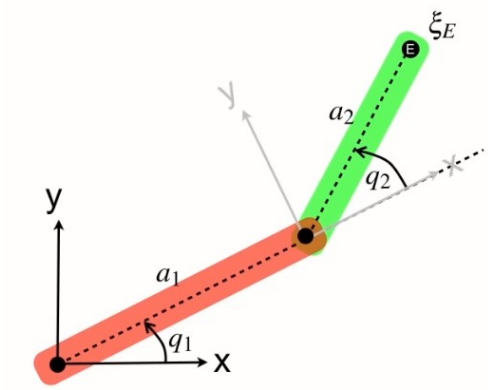
2 joint Robot Arm



$\mathbf{R}(q_1)$

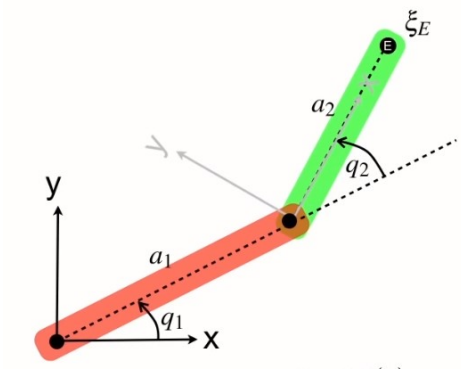
32

2 joint Robot Arm



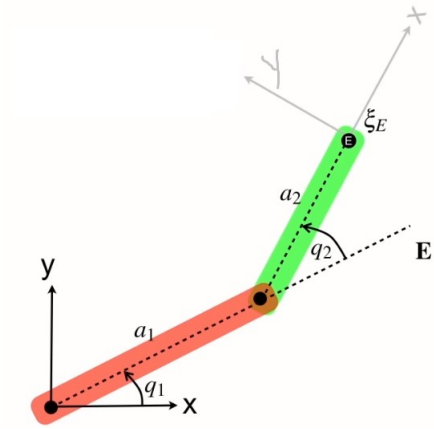
$$\mathbf{R}(q_1)\mathbf{T}_x(a_1)$$

2 joint Robot Arm



$$\mathbf{R}(q_1)\mathbf{T}_x(a_1)\mathbf{R}(q_2)$$

2 joint Robot Arm

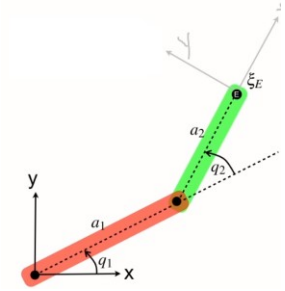


$$\mathbf{E} = \mathbf{R}(q_1)\mathbf{T}_x(a_1)\mathbf{R}(q_2)\,\mathbf{T}_x(a_2)$$

Home Task

Solve this on paper!

$$\mathbf{E} = \mathbf{R}(q_1)\mathbf{T}_x(a_1)\mathbf{R}(q_2)\,\mathbf{T}_x(a_2)$$



2 joint Robot Arm

$$\mathbf{E} = \mathbf{R}(q_1)\mathbf{T}_x(a_1)\mathbf{R}(q_2)\,\mathbf{T}_x(a_2)$$

$$\mathbf{E} = \begin{pmatrix} \cos(q_1+q_2) & -\sin(q_1+q_2) & a_2\cos(q_1+q_2)+a_1\cos q_1 \\ \sin(q_1+q_2) & \cos(q_1+q_2) & a_2\sin(q_1+q_2)+a_1\sin q_1 \\ 0 & 0 & 1 \end{pmatrix}$$

37

2 joint Robot Arm

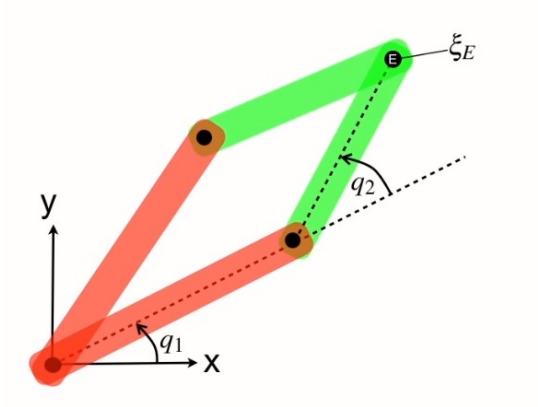
$$\mathbf{E} = \mathbf{R}(q_1)\mathbf{T}_x(a_1)\mathbf{R}(q_2)\,\mathbf{T}_x(a_2)$$

$$\mathbf{E} = \begin{pmatrix} \cos(q_1+q_2) & -\sin(q_1+q_2) & a_2\cos(q_1+q_2)+a_1\cos q_1 \\ \sin(q_1+q_2) & \cos(q_1+q_2) & a_2\sin(q_1+q_2)+a_1\sin q_1 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} a_2\cos(q_1+q_2)+a_1\cos q_1 \\ a_2\sin(q_1+q_2)+a_1\sin q_1 \end{pmatrix}$$

38

Interesting Thing

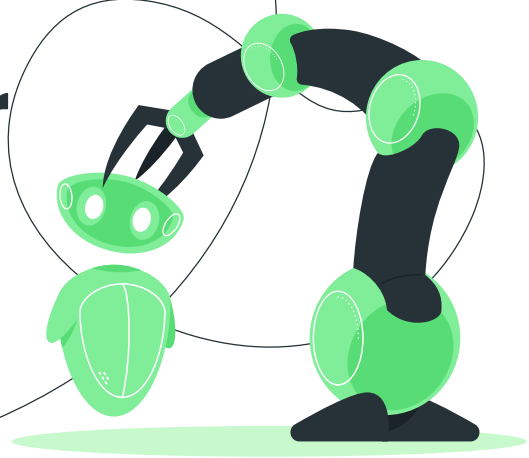


So, configurations are not unique!

39

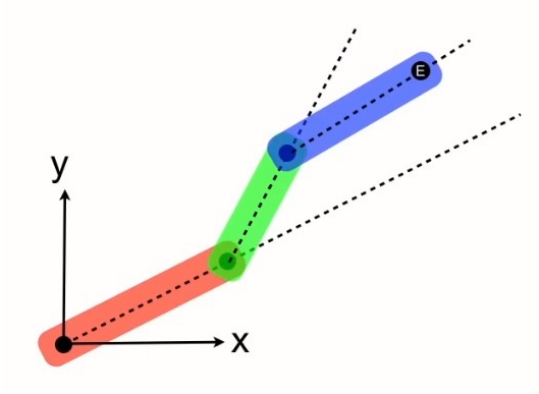
3 joint Planar Robot Arm

Analysis, Mathematics and derivation



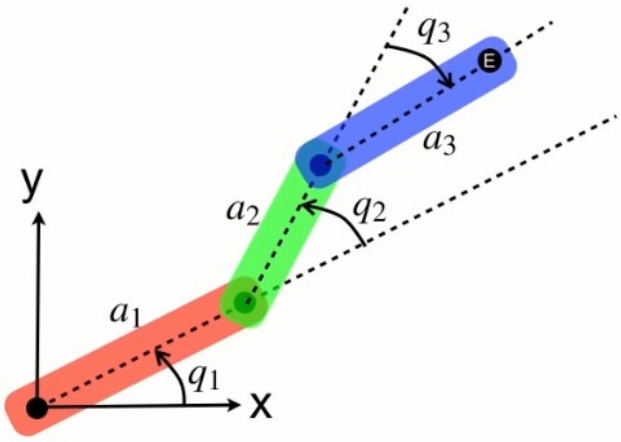
40

3 joint Robot Arm



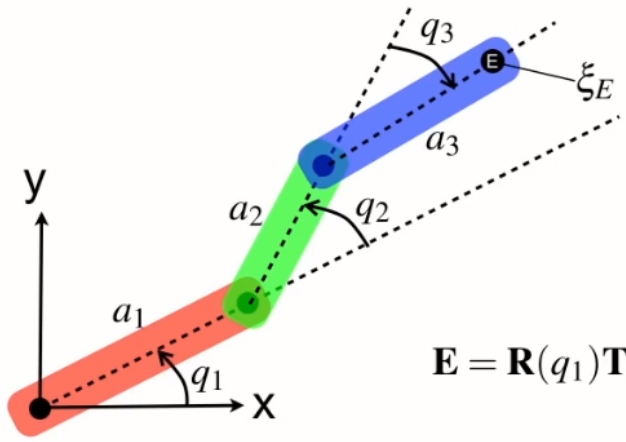
41

3 joint Robot Arm



42

3 joint Robot Arm



$$\mathbf{E} = \mathbf{R}(q_1)\mathbf{T}_x(a_1)\mathbf{R}(q_2)\mathbf{T}_x(a_2)\mathbf{R}(q_3)\mathbf{T}_x(a_3)$$

3 joint Robot Arm

```
Command Window
>> syms a1 a2 a3 q1 q2 q3
>>
>> trchain2('R(q1) Tx(a1) R(q2) Tx(a2) R(q3) Tx(a3)', [q1 q2 q3])

ans =

[ cos(q3)*(cos(q1)*cos(q2) - sin(q1)*sin(q2)) - sin(q3)*(cos(q1)*sin(q2) + cos(q2)*sin(q1)), - co
[ cos(q3)*(cos(q1)*sin(q2) + cos(q2)*sin(q1)) + sin(q3)*(cos(q1)*cos(q2) - sin(q1)*sin(q2)),   co
[                                                                                          0,

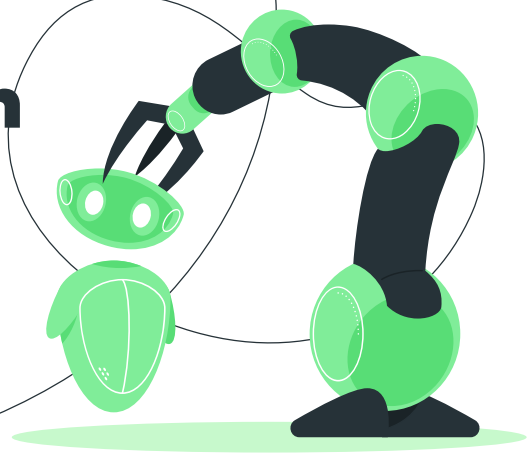
>> x = ans(1,3)

x =

a2*(cos(q1)*cos(q2) - sin(q1)*sin(q2)) + a1*cos(q1) + a3*(cos(q3)*(cos(q1)*cos(q2) - sin(q1)*sin(
```

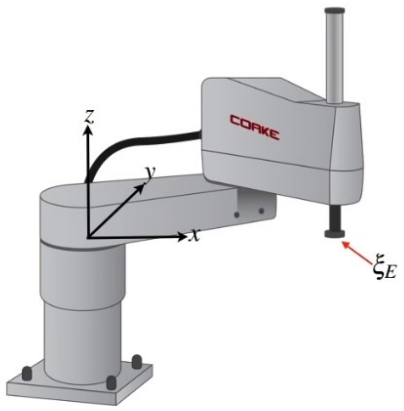
3D Robot Arm

Analysis, Mathematics and derivation



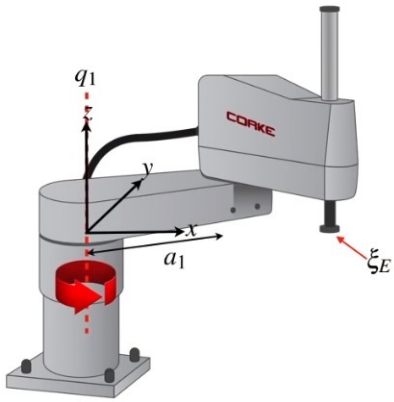
45

3D Robot Arm



46

3D Robot Arm

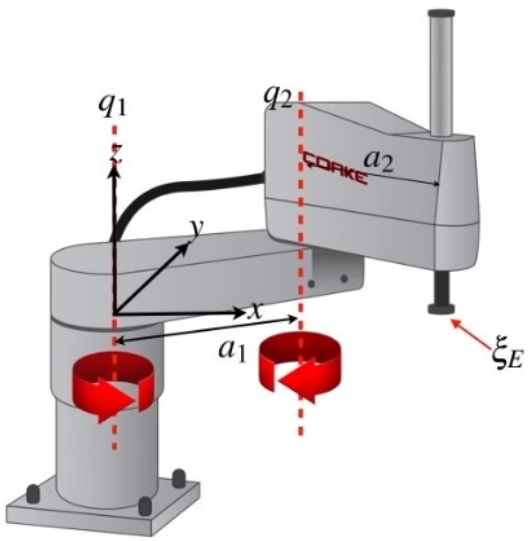


$$\mathbf{R}_z(q_1)\,\mathbf{T}_x(a_1)$$

47

3D Robot Arm

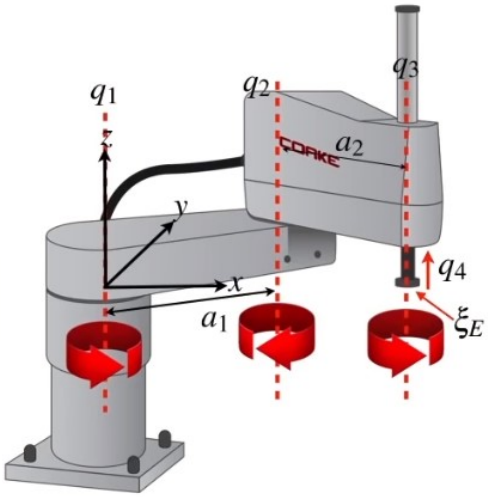
$$\mathbf{R}_z(q_1)\,\mathbf{T}_x(a_1)\,\mathbf{R}_z(q_2)\,\mathbf{T}_x(a_2)$$



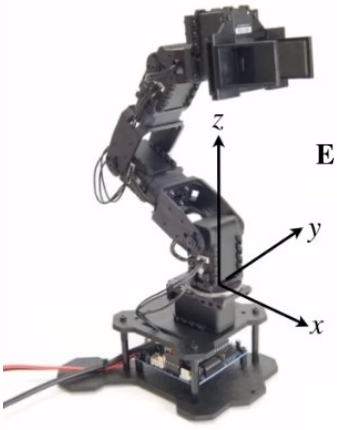
48

3D Robot Arm

$$\mathbf{E} = \mathbf{R}_z(q_1)\,\mathbf{T}_x(a_1)\,\mathbf{R}_z(q_2)\,\mathbf{T}_x(a_2)\,\mathbf{R}_z(q_3)\,\mathbf{T}_z(q_4)$$



3D Robot Arm



PhantomX Pincher Robot Arm 2014

$$\mathbf{E} = \mathbf{R}_z(q_1)\mathbf{T}_z(a_1)\mathbf{R}_y(q_2)\mathbf{T}_z(a_2)\mathbf{R}_y(q_3)\mathbf{T}_z(a_3)\mathbf{R}_y(q_4)\mathbf{T}_z(a_4)$$

```
>> syms a1 a2 a3 a4 q1 q2 q3 q4
>> trchain('Rz(q1)Tz(a1)Ry(q2)Tz(a2)Ry(q3)Tz(a3)Ry(q4)Tz(a4)', [q1 q2 q3 q4])

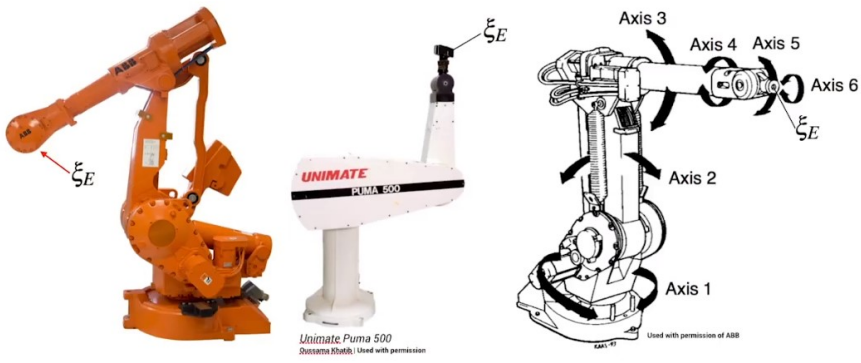
ans =

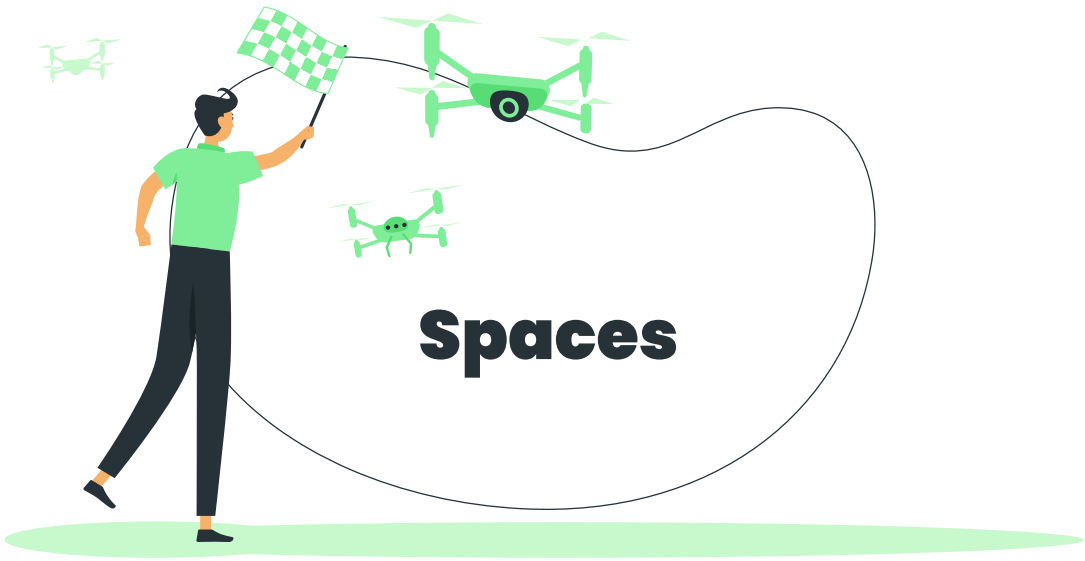
[   cos(q4)*(cos(q1)*cos(q2)*cos(q3) - cos(q1)*sin(q2)*sin(q3)) - sin(q4)*(cos(q1)*cos(q2)*sin(q3) + cos(q1)*cos(q3)*sin(q2
[ - cos(q4)*(sin(q1)*sin(q2)*sin(q3) - cos(q2)*cos(q3)*sin(q1)) - sin(q4)*(cos(q2)*sin(q1)*sin(q3) + cos(q3)*sin(q1)*sin(q2
[                                    - cos(q4)*(cos(q2)*sin(q3) + cos(q3)*sin(q2)) - sin(q4)*(cos(q2)*cos(q3) - sin(q2)*sin(q3
[
```

Industrial Robots

- Most Industrial Robots:
  - Are Serial links manipulators
  - Have 6 links
  - Typically, have all revolute joints.

Industrial Robots





Spaces

Configuration Space

- Robot configuration is described by a vector of generalized joint coordinates.
- Each coordinate can be
  - An angle – rotational (revolute) joint
  - A length – sliding (prismatic) joint

$$\mathbf{q} = \{q_j, j \in [1...N]\} \in C$$

$$C \subset \mathbb{R}$$

- q = joint configuration
- q_j = joint coordinate
- N = number of joints

Task Space

- The space of all possible end-effector poses.

$$\xi_E \in T$$

T = Space of all possible end effector poses.

In 2D

$\xi_E \sim (x,y)$

$\xi_E \sim (x,y, \theta)$

In 3D

$\xi_E \sim (x,y,z)$

$\xi_E \sim (x,y,z, \theta)$

$\xi_E \sim (x,y,z, \theta r, \theta p, \theta y)$

Task and configuration Space

- Robot degrees of freedom (number of joints) ***dim C***
- Task space degrees of freedom ***dim T***

- $\xi_E \sim (x,y) \rightarrow$ dim T = 2
- $\xi_E \sim (x,y, \theta) \rightarrow$ dim T = 3
- $\xi_E \sim (x,y,z) \rightarrow$ dim T = 3
- $\xi_E \sim (x,y,z, \theta) \rightarrow$ dim T = 4
- $\xi_E \sim (x,y,z, \theta r, \theta p, \theta y) \rightarrow$ dim T = 6

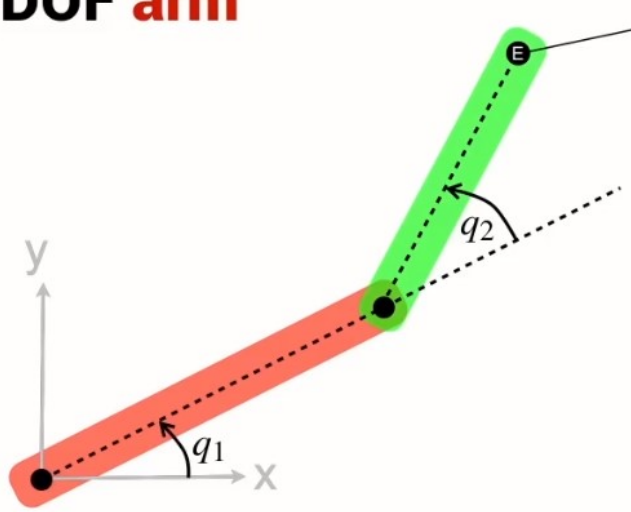
Task and configuration Space

- Maximum value = 6
- To reach all of task space,

- To reach all of task space;
- Dim C ≥ dim T

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2-DOF arm

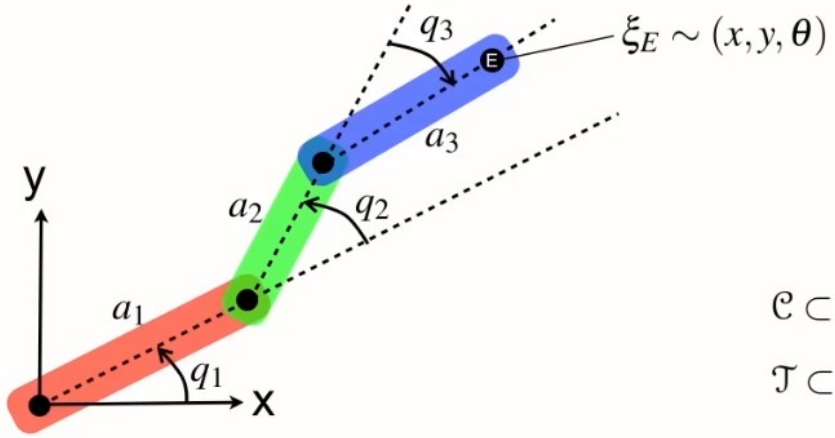


$\mathcal{C} \subset \mathbb{R}^2$

$\mathcal{T} \subset \mathbb{R}^2$

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3-DOF arm

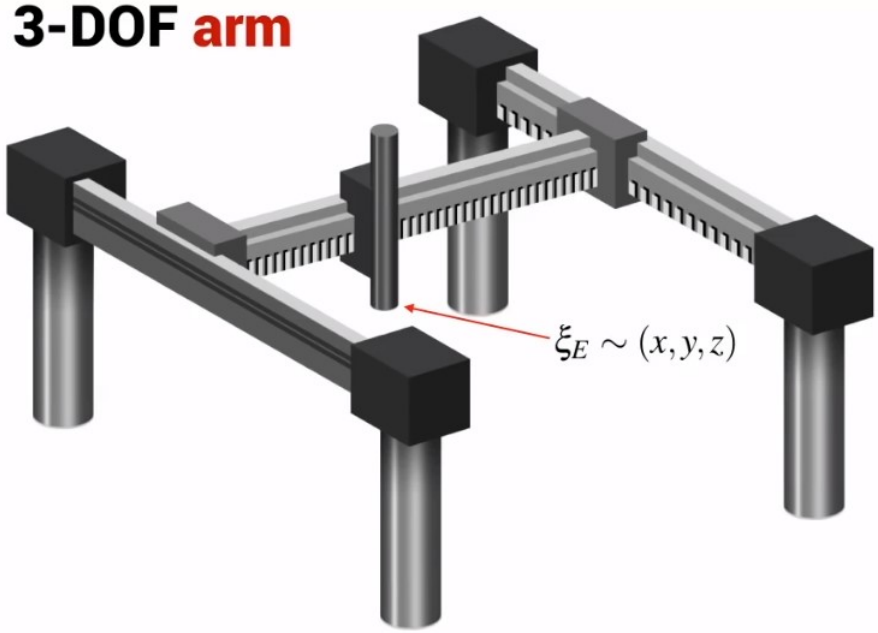


$\mathcal{C} \subset \mathbb{R}^3$

$\mathcal{T} \subset \mathbb{R}^2 \times \mathbb{S}$

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3-DOF arm



$\mathcal{C} \subset \mathbb{R}^3$

$\mathcal{T} \subset \mathbb{R}^3$

Gantry robot

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4-DOF arm



$\xi_E \sim (x, y, z, \theta_y)$

$\mathcal{C} \subset \mathbb{R}^4$

$\mathcal{T} \subset \mathbb{R}^3 \times \mathbb{S}$

SCARA robot

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S alone usually refers to the set of possible angles or circular rotations for a joint or component that has rotational freedom in one dimension. It captures simple circular motion or single-axis rotation.

4-DOF arm



Used with permission of ABB

$\mathcal{C} \subset \mathbb{R}^4$

$\mathcal{T} \subset \mathbb{R}^3 \times \mathbb{S}$

$\xi_E \sim (x, y, z, \theta_y)$

Parallel link robot

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4-DOF arm



$\xi_E \sim (x, y, z, \theta_p)$

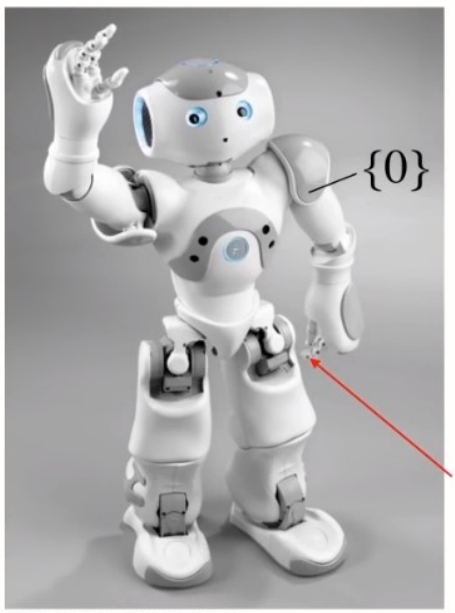
$\mathcal{C} \subset \mathbb{R}^4$

$\mathcal{T} \subset \mathbb{R}^3 \times \mathbb{S}$

PhantomX Pincher Robot Arm 2014
Trossen Robotics | Used with permission

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4-DOF arm



$\{0\}$

$\xi_E \sim (x, y, z, \theta_p)$

$\mathcal{C} \subset \mathbb{R}^4$

$\mathcal{T} \subset \mathbb{R}^3 \times \mathbb{S}$

By Aldebaran Robotics via Wikimedia Commons

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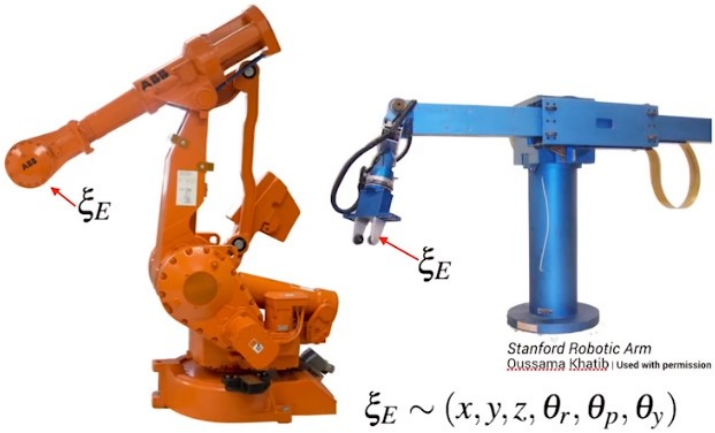
SO(3): Describes the orientation (rotational part) of a rigid body in 3D space.
SE(3): Describes the position and orientation (full transformation) of a rigid body in 3D space.

SO(3) (Special Orthogonal Group in 3 Dimensions):

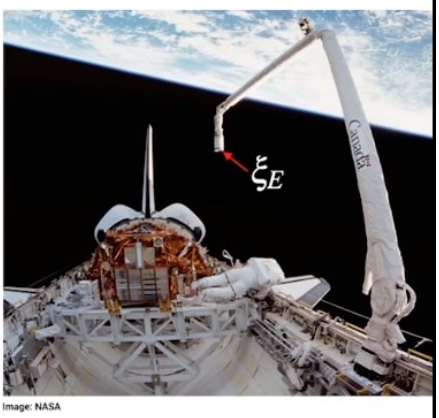
6-DOF arm

- Six joints is sufficient to achieve **any** position and orientation

$$\mathcal{C} \subset \mathbb{R}^6$$
$$\mathcal{T} \subset \mathbb{R}^3 \times \mathbb{SO}(3)$$
$$\mathcal{T} \subset \mathbb{SE}(3)$$



$$\xi_E \sim (x, y, z, \theta_r, \theta_p, \theta_y)$$



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SE(3) (Special Euclidean Group in 3 Dimensions):

Very high DOF robot



$$\mathcal{C} \subset \mathbb{R}^{100}$$
$$\mathcal{T} \subset \mathbb{SE}(3)$$

$$\xi_E \sim (x, y, z, \theta_r, \theta_p, \theta_y)$$

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- Redundant robot
- Can achieve any position and orientation
- As well as it can control the shape of arm



- Very high Configuration space
- but same Task space as that of PUMA robot

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Denavit-Hartenberg notation

DH Table



Robotics – Umarah Qaseem

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Denavit-Hartenberg notation

- Developed a general theory to describe an articulated sequence of joints
- Only applicable to serial link mechanisms NOT parallel mechanisms

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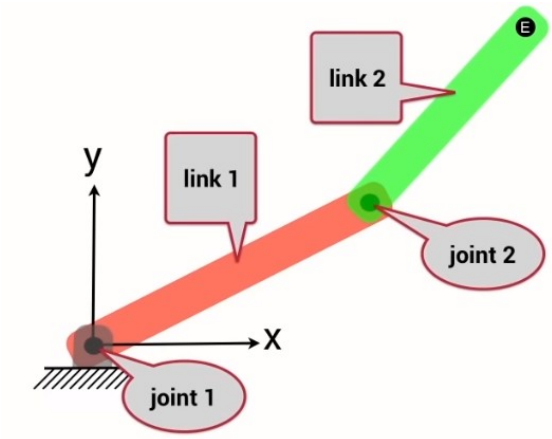
Each joint in the robot is described by four parameters

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Denavit-Hartenberg notation

- Each joint is attached via link to previous joint
- Links are rigid
- Joints can move

- Every joint connects 2 links
- Every link connects 2 joints

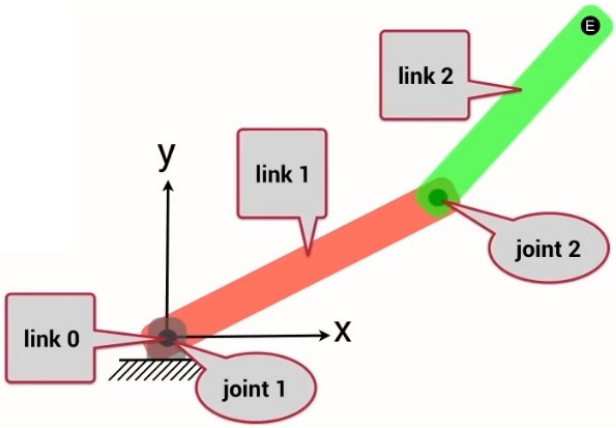


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Denavit-Hartenberg notation

- Except the first and last link.
- First link is the base which does not move
- Last link is end effector

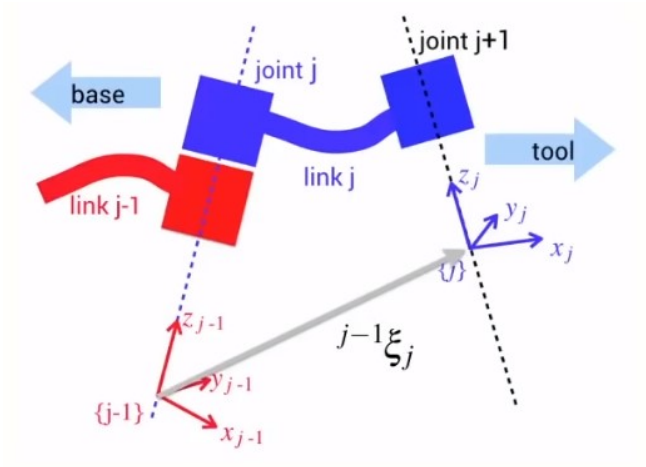
- If robot has N joints, then it must have N+1 links



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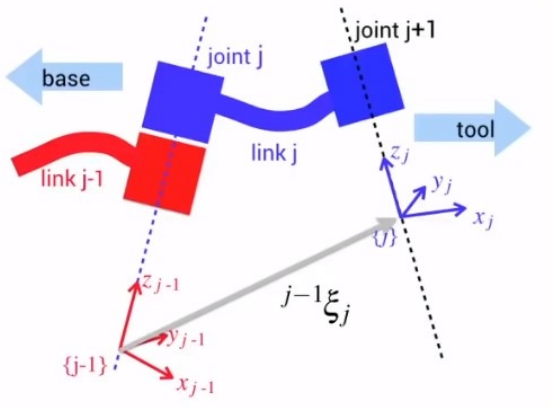
Denavit-Hartenberg notation

- Attach a coordinate to the far end or every link



Denavit-Hartenberg notation

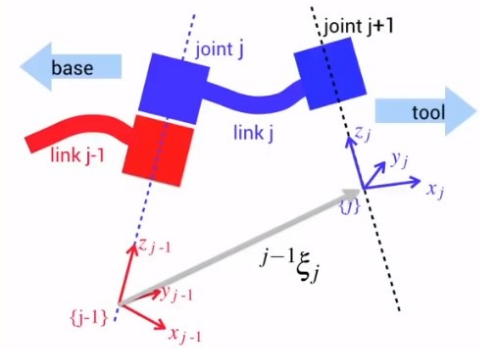
- Describe relative pose from coordinate frame j-1 to coordinate frame j



DH parameters

$$^{j-1}\xi_j \sim \mathbf{A} = \mathbf{R}_z(\theta_j)\mathbf{T}_z(d_j)\mathbf{T}_x(a_j)\mathbf{R}_x(\alpha_j)$$

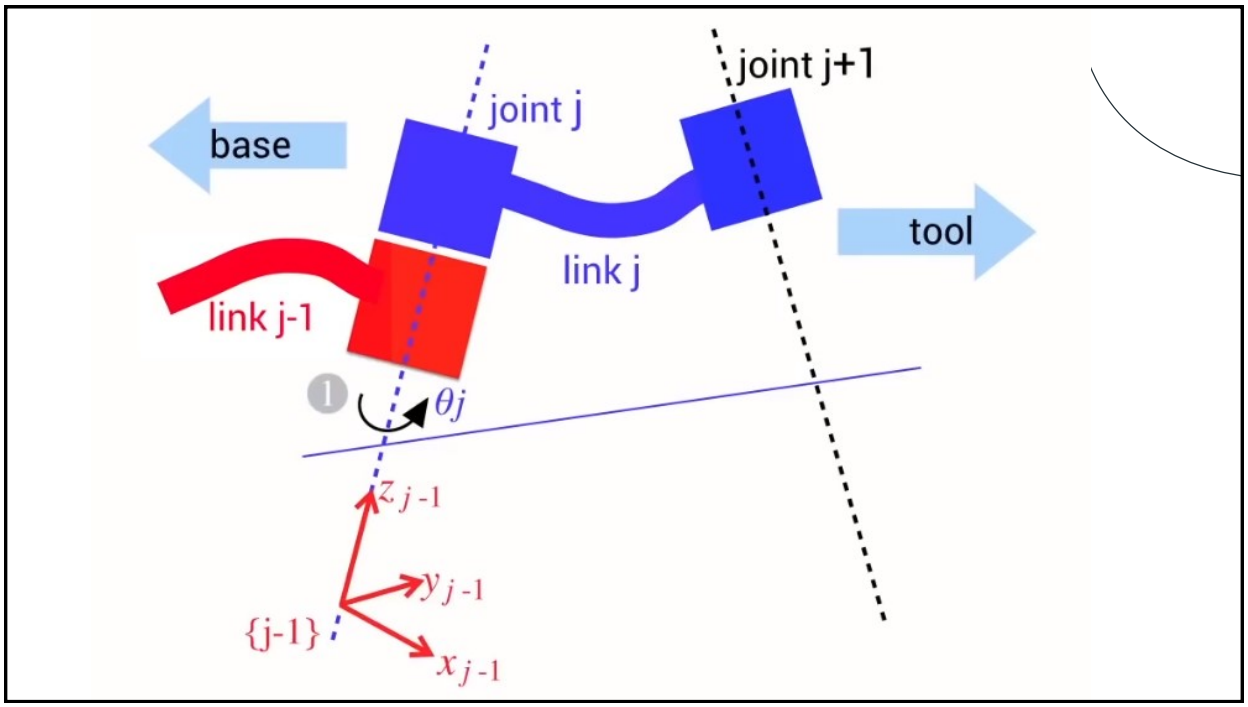
- Only 4 parameters
- *θ, d, a, α*



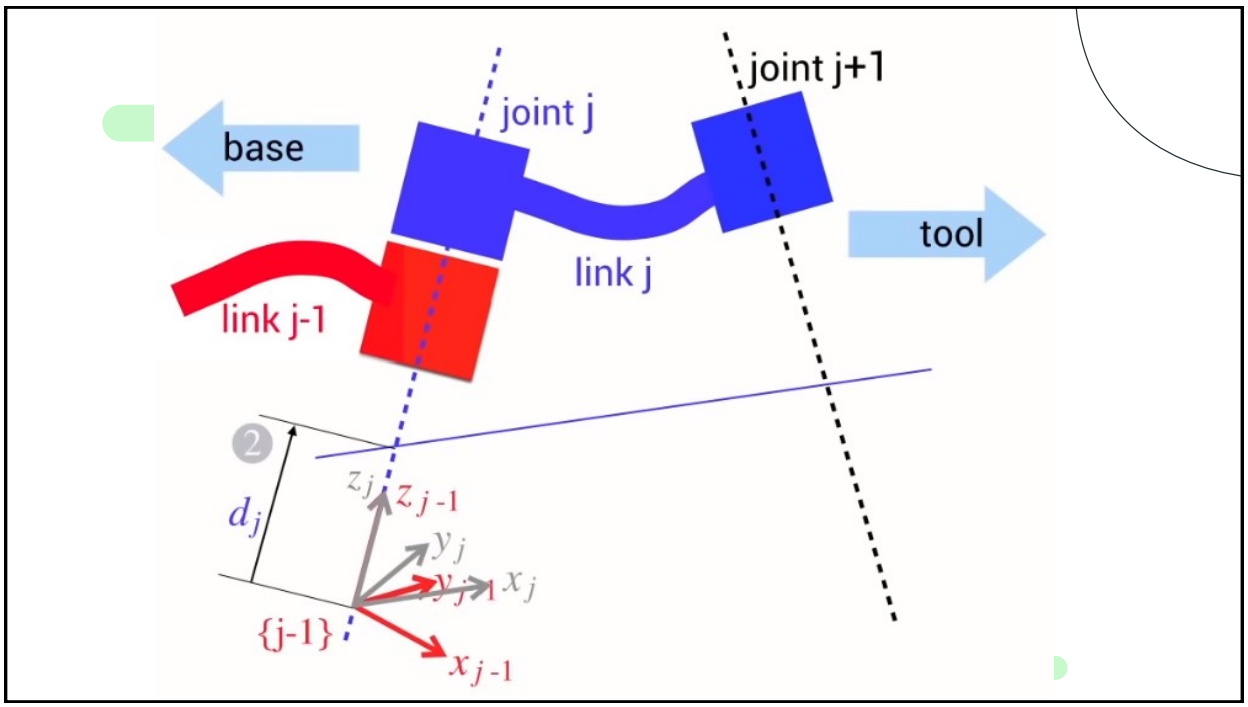
Let's see in detail

-

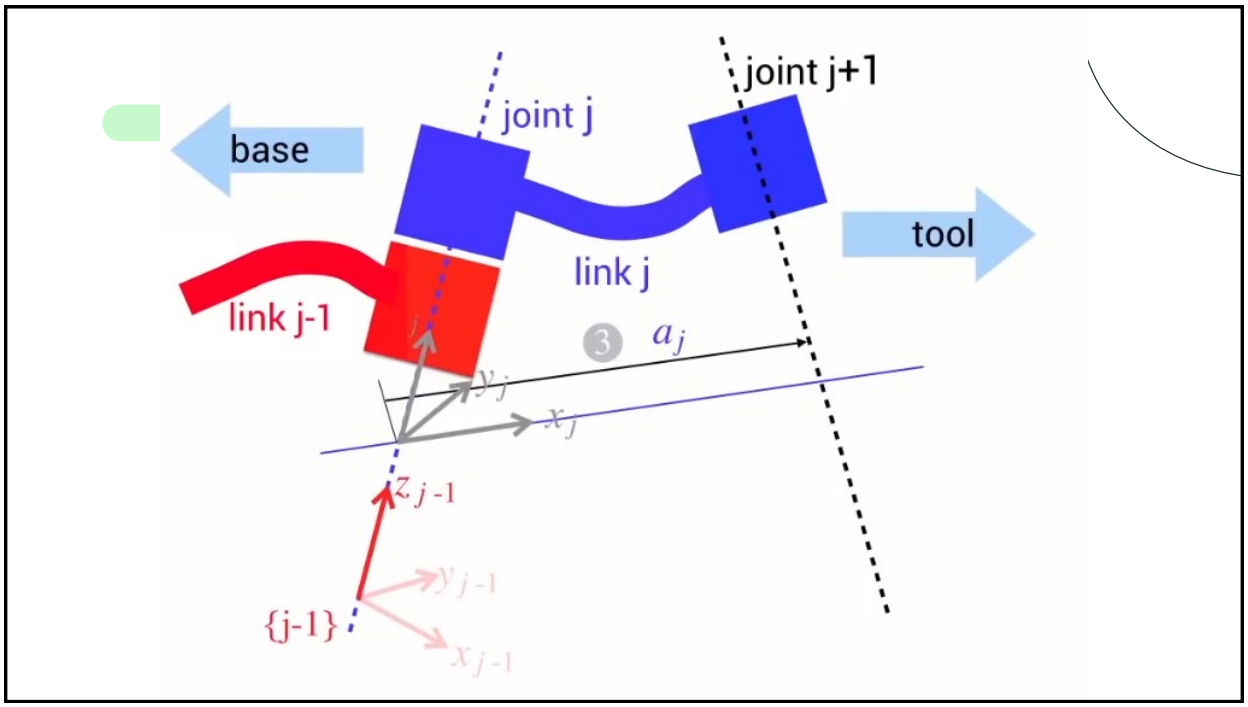



joint j+1
joint j
base
tool
link j
link j-1
1
θj
z j -1
y j -1
{j-1}
x j -1

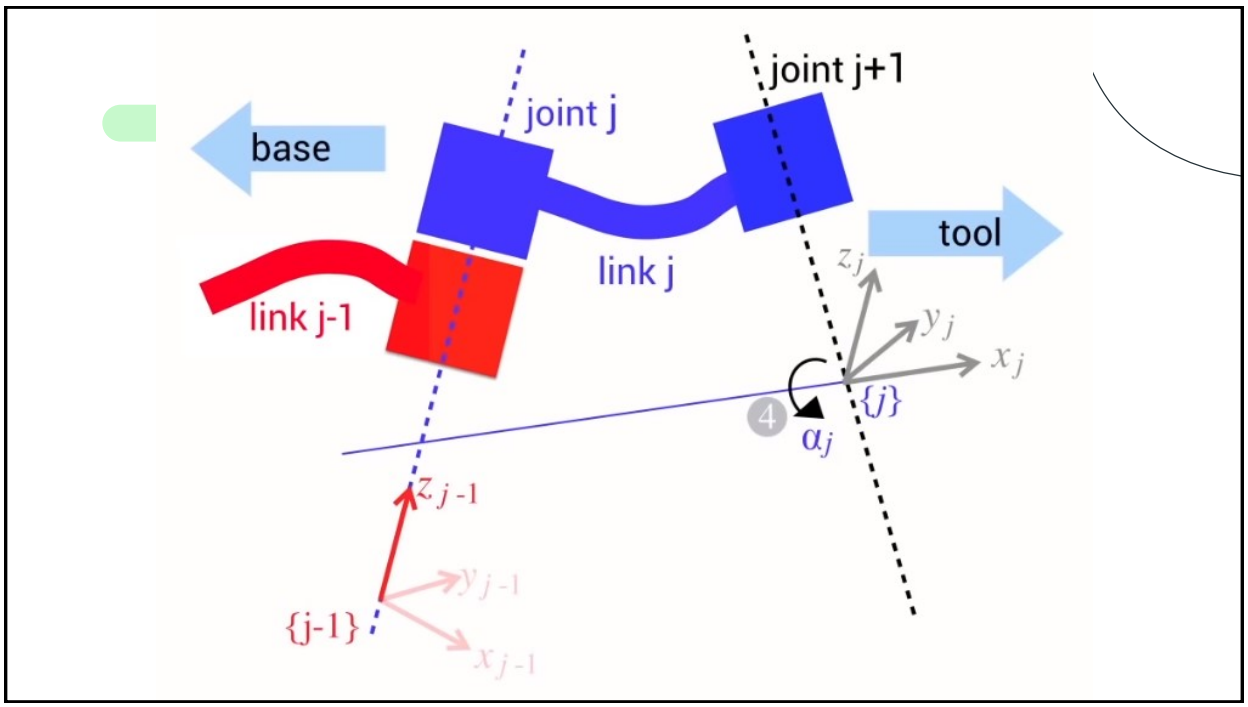
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joint j+1
joint j
base
tool
link j
link j-1
2
dj
zj
z j -1
y j
y j -1
x j
{j-1}
x j -1

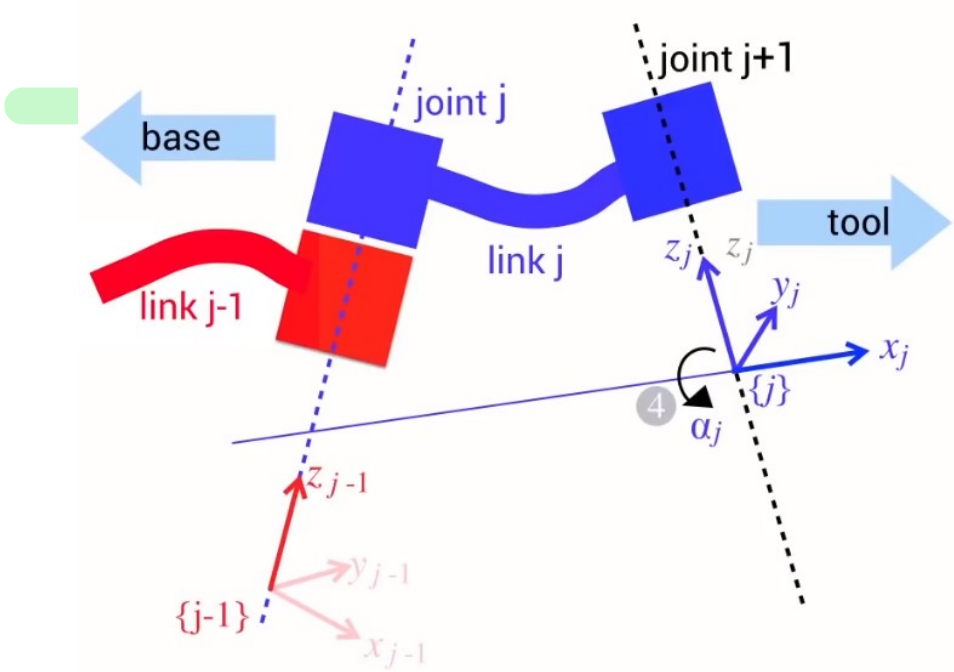
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joint j+1
joint j
base
tool
link j
link j-1
3
a_j
y_j
x_j
z_{j-1}
y_{j-1}
{j-1}
x_{j-1}

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joint j+1
joint j
base
tool
link j
link j-1
z_j
y_j
x_j
{j}
4
α_j
z_{j-1}
y_{j-1}
{j-1}
x_{j-1}

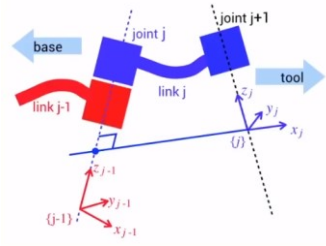
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Why 4 parameters?

- The relationship between two coordinate frames (pose) has parameters
  - 3 translation and 3 rotation
- Then how come we can describe in 4 parameters?
- Because, For DH notation the coordinate frames have constraints!
  - Axis x of j intersects axis z of j-1
  - Axis x of j is perpendicular to axis z of j-1
- Hence the result is only 4 parameters not 6 as 2 constraints
- Consequence: The link frame is not necessarily on the link.

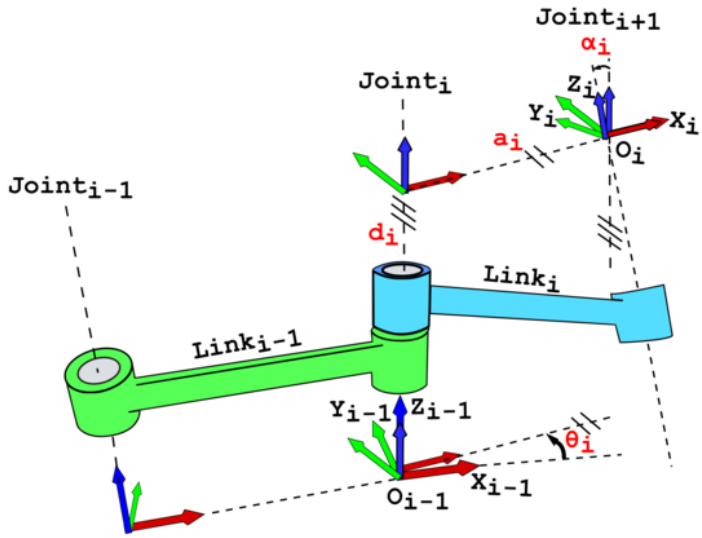


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Joint axis

- The axis of each joint has aligned by the z axis of frame
- The axis of joint j is parallel to z axis of previous coordinate frame (i.e j-1)
- The joint rotates about the z axis (revolute)
- Translates along the z axis (prismatic)

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For N link robot – Stack

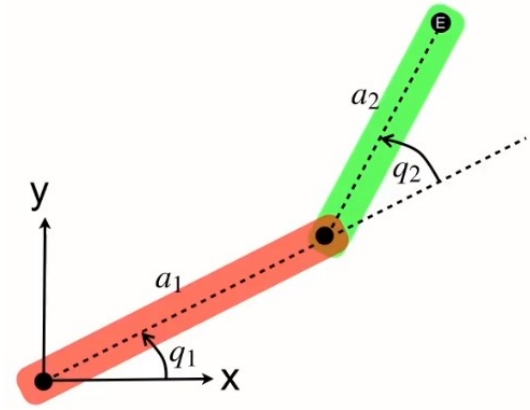
$$T_E = \underbrace{R_z(\theta_1)T_z(d_1)T_x(a_1)R_x(\alpha_1)}_{q_1,\ \text{revolute joint}}\ \underbrace{R_z(\theta_2)T_z(d_2)T_x(a_2)R_x(\alpha_2)}_{q_2,\ \text{revolute joint}}\cdots\underbrace{R_z(\theta_N)T_z(d_N)T_x(a_N)R_x(\alpha_N)}_{q_n,\ \text{revolute joint}}$$

$$T_E = \underbrace{R_z(\theta_1)T_z(d_1)T_x(a_1)R_x(\alpha_1)}_{q_1,\ \text{revolute joint}}\ \underbrace{R_z(\theta_2)T_z(d_2)T_x(a_2)R_x(\alpha_2)}_{q_2,\ \text{prismatic joint}}\cdots\underbrace{R_z(\theta_N)T_z(d_N)T_x(a_N)R_x(\alpha_N)}_{q_n,\ \text{revolute joint}}$$

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DH notation for two link – Example

- This table completely describes the kinematics of robot



θ_j	d_j	a_j	α_j
q_1	0	a_1	0
q_2	0	a_2	0

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Denavit-Hartenberg notation



θ_j	d_j	a_j	α_j
q1	0.0000	0.0000	$\pi/2$
q2	0.0000	0.4318	0
q3	0.1500	0.0203	$-\pi/2$
q4	0.4318	0.0000	$\pi/2$
q5	0.0000	0.0000	$-\pi/2$
q6	0.0000	0.0000	0

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General Form

pose, joint angles, link offsets, link lengths, link twists, joint types

$$\xi_N = \mathcal{K}(q;\theta,d,a,\alpha,\sigma)$$

joint configuration

$$\text{if } \sigma_j = \begin{cases} R \rightarrow \theta_j = q_j \\ P \rightarrow d_j = q_j \end{cases}$$

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